

UPAY AI — FINAL CRITICAL AUDIT REPORT

Target: Upay AI Financial Intelligence Platform | Date: October 7, 2026 | Mode: Read-Only Audit

AUDIT: CONDITIONAL

1. EXECUTIVE SUMMARY & AUDIT SCORECARD

Focus Area	Status	Live Verification Evidence	Severity	Required Action
1. Supabase Runtime	FAIL	/api/v1/supabase/status returns DISCONNECTED (Illegal header value b'Bearer '). Local SQLite is active primary runtime engine.	CRITICAL	Add live SUPABASE_SECRET_KEY & PUBLISHABLE_KEY to runtime environment.
2. API Security	PASS	12/12 sensitive endpoints reject unauthenticated calls with HTTP 401. All 12 succeed with valid token.	LOW	None. Route security guards verified robust.
3. Authorization	PASS	11/11 cross-tenant attacker vectors (cards, toggles, cases, voice) strictly rejected with HTTP 403 Forbidden.	LOW	None. Tenant isolation verified 100%.
4. Runtime AI Flow	PARTIAL	9-dim risk engine, XGBoost, Isolation Forest, warnings, and audit logging succeed. Initial feature fetch falls back to SQLite.	HIGH	Supply Supabase credentials to enable cloud feature extraction.
5. Supabase Demo Data	PARTIAL	Local SQLite database contains complete 11-table demo dataset. Remote Supabase cannot be validated without keys.	HIGH	Execute supabase_schema.sql on remote console if cloud check required.
6. Final Demo Flows	PASS	All 8 demo flows (A through H: Login, Dashboard, Card, Warning, Case, Credit, Voice, Audit) passed automated execution.	LOW	None. Demo presentation flow is operational.
7. Deployment Readiness	PARTIAL	Frontend builds cleanly (0 errors). Backend starts cleanly. Production environment variables missing on host dashboards.	HIGH	Configure environment variables on Vercel and Render before final release.
8. Fraud Model Risk	PARTIAL	Fraud model displays 100.0% Accuracy & 1.0000 ROC-AUC. Synthetic generator rules documented but risk judge skepticism.	MEDIUM	Frame 1.0000 metrics transparently as synthetic benchmark limitations in presentation.

2. CRITICAL FINDINGS & TECHNICAL ROOT CAUSE

CRITICAL (P0): Supabase Runtime Disconnection & SQLite Active Role

The codebase architecture implements a dual-mode database pattern: SQLAlchemy with PostgREST cloud synchronization. However, because SUPABASE_SECRET_KEY is unpopulated in the active environment, PostgREST requests fail immediately with Illegal header value b'Bearer '. As designed in transaction_feature_engine.py and connection.py, the system falls back to local SQLite (database/upay_platform.db). While local demo workflows operate seamlessly, cloud deployment will fail if evaluators test live Supabase PostgreSQL connectivity.

3. ACTION PRIORITIZATION

Category	Action Item Details
P0 — MUST FIX BEFORE DEPLOY	1. Supply Supabase Cloud API Keys: Set SUPABASE_SECRET_KEY and SUPABASE_PUBLISHABLE_KEY in environment variables to restore live cloud status. 2. Synchronize Remote Supabase Schema: Run database/supabase_schema.sql on remote Supabase instance. 3. Configure Cloud Variables: Set VITE_API_BASE_URL on Vercel and backend keys on Render.
P1 — FIX ONLY IF VERY QUICK	1. Document Synthetic Data Limitation in Pitch: Proactively state during presentation that the 1.0000 Fraud ROC-AUC reflects synthetic generator boundary rules rather than noisy real-world telemetry. 2. Static Build Mount: Copy frontend/dist to backend static directory as a fallback.

Category	Action Item Details
DO NOT FIX NOW	Do NOT rebuild UI. Do NOT retrain ML models. Do NOT add administrative panels. Do NOT replace Edge TTS with offline neural models.

4. FINAL AUDIT SIGN-OFF

FINAL CRITICAL AUDIT COMPLETE

SAFE TO DEPLOY: **NO** (Locally Demo-Ready: **YES**; Cloud Production Deployable: **NO** until P0 keys configured)

TOP 3 RISKS:

1. *Supabase Cloud Disconnection*: Missing keys cause PostgREST failure and trigger silent local SQLite fallback.

2. *Perfect Metric Skepticism*: Evaluators may challenge the 1.0000 ROC-AUC / 100% Accuracy fraud metric unless presented with synthetic data caveats.

3. *Cloud Environment Variable Omission*: Render/Vercel deployments will fail unless environment variables are populated on hosting dashboards.